MIL-P-17959(NOrd)
6 April 1954

MILITARY SPECIFICATION

POWDER, PROPELLANT, BALLISTICS OF;
FOR 3"/50 CALIBER GUN

1. SCOPE

1.1 Scope. - This specification contains the methods and criteria for acceptance testing of 3"/50 caliber gun production powders with respect to ballistic properties. Ballistic testing according to this specification is in addition to and subsequent to chemical and physical requirements for the manufacture of 3"/50 caliber gun propellants which are specified according to propellant type in accordance with the specifications listed in 2.1.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND PUBLICATIONS

2.1 SPECIFICATIONS. - The following specifications refer to chemical and physical requirements pertaining to the manufacture of 3"/50 caliber gun propellant powders:

JAN-P-231	- Powder, Smokeless, Pyrocellulose
JAN-P-309	- Powder, Propellant, Cannon, M-1, M-6, and M-14
MIL-P-17449(NOrd)	- Powder, Propellant, Cordite N

2.2 DRAWINGS. - The following list of drawings forms a part of this specification:

SK No.	90564	Mk 21-0	3"/50 Caliber gun
"	90566	Mk 21-1	"
"	91143	Mk 21-2	"
"	62199	Mk 22-0	"
"	107710	Mk 22-1	"
"	107715	Mk 22-2	"
"	107716	Mk 22-3	"
"	167534	Mk 22-4	"
"	167535	Mk 22-5	"
LD No.	167583	Mk 22-6	"
"	167584	Mk 22-7	"
"	262340	Mk 22-8	"
"	262382	Mk 22-9	"
"	255506	Mk 33	Projectile

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3. REQUIREMENTS

3.1 Charges, pressures, and velocities. - Weight of charge, muzzle velocity, and pressure results obtained with the test powder according to Section 4 must conform to the requirements listed below for the type of powder involved, using Mk 33 projectiles inert loaded to 13.00 pounds. (It is to be noted that charge weight and maximum pressure limits correspond to values adjusted for deviations of reference propellant performance from established standards.)

PROPELLANT TYPE	MASTER TYPE	CHARGE* WEIGHT Limits Min.	Max.	PRESSURE** Min.	Max.	UNIFORMITY*** Pressure	Velocity
Cordite N	Cordite N	4.00	4.45	15.0	17.0	0.4	13
M6 and Pyro Flashing	M-6	3.85	4.30	15.0	17.0	0.4	13
M6 and Pyro Flashless	M-6	3.90	4.35	15.0	17.0	0.4	13

* Pounds of powder required to match master powder velocity.
** Long tons per square inch divided by 1.2 (copper crusher pressure).
*** Mean deviation, i.e. average difference between mean and individual results for five rounds fired according to Condition 6 of 4.4. Pressure units in long tons per square inch divided by 1.2 and velocity units in feet per second (f/s).

The criteria listed above are to be applied to the mean of the results obtained with each of the two guns. Exception: When testing pyro flashing or flashless powders or M-6 flashless powders against the M-6 flashing master powder, two guns at different stages of wear must be used. The two guns should be chosen so that the master powder mean velocities differ by at least 30 f/s. In such cases, the average pressure and uniformity results are to be used, but the final charge is to be determined by means of the formula of 4.5.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Definitions

4.1.1 Master powder. - The master powder is a specifically designated lot of powder with well established and satisfactory ballistic

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properties which is used as a reference standard for all powders of its type. Two master powders are used in testing 3"/50 caliber gun propellant powders. A cordite N master powder is used in testing cordite N propellants, and an M6 flashing master powder is used in testing all M6 and pyro propellants. (A quantity of the master powder will be furnished for ballistic testing. Ballistic properties of the master powder, such as assigned maximum breech pressure, will be furnished with the powder.)

4.1.2 Pressure and velocity of the day. - The mean maximum breech pressure and muzzle velocity of the master powder in the firing in a gun on a particular day are known as the pressure and velocity of the day. A five round mean result is usually accepted as defining this.

4.1.3 Matched powder method of powder proof. - Assignments of a charge to a test powder which will yield the velocity of the day as obtained with the master powder is termed the matched powder method.

4.1.4 Maximum packing depth (MPD). - MPD is the distance, to the nearest tenth of an inch, from the case mouth to the upper level of any given powder charge in the case under the following conditions:

a. The charge is poured from a distance thirty inches above the case mouth.

b. The charge is poured steadily over a time interval of one minute.

c. The outside of the case is lightly tapped with a soft mallet adjacent to the rising level of the powder during pouring.

MPD varies with powder lot and charge weight.

4.1.5 Production packing depth (PPD). - For a fixed charge of powder, PPD is defined as the MPD for that charge less 0.4 inch. It is a powder level which can be readily attained in production assembly of rounds.

4.1.6 Conditioning round. - The first round fired in a gun during a firing program, or a round interposed between two rounds assembled with different types of powder charges, is called a conditioning round. Conditioning rounds are utilized between rounds assembled with any two of the following: Cordite N, pyro flashing, pyro flashless, M6 flashing, and M6 flashless powder. Conditioning rounds may have unusual

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ballistics, and ballistic results obtained with them are disregarded in powder proof. Conditioning rounds are also used to obviate effects on a gun because of firing at high pressures. For example, conditioning rounds are fired when a charge is significantly lighter than its predecessor.

4.1.7 Assigned master powder pressure. - This is the pressure established as a mean new gun pressure for the master powder. The difference of the assigned pressure minus the pressure of the day for the master powder is added algebraically to all test powder pressures, and it is this corrected test powder pressure which is subject to the pressure limit specifications.

4.2 Preparation of charges. - Propellants to be fired in ballistic testing for determination of charge weight shall be conditioned by being stored at a constant temperature of 90 degrees Fahrenheit (F) for at least 7 days immediately prior to firing. It is important that the powder temperature be maintained as near to 90°F as possible until fired. A charge and powder level will be specified for the master powder when the sample is furnished, and every round fired with the master powder in ballistic testing is to be loaded to these specifications. A tolerance of ± 0.1 inch is permitted on the powder level. There is no permissable tolerance on the charge weights of the master powder. (Particular care is required in conducting ballistic testing. Every effort should be made to eliminate unnecessary variables which might affect muzzle velocity. For example, it is important that projectiles be inert loaded to specified weight.) PPD as a function of charge weight must be determined for each lot of test powder. This is accomplished by pouring five charges each at weights of 3.30 and 4.40 pounds under MPD conditions, drawing a straight line through the average MPD levels thus obtained and drawing a PPD line by subtracting a constant 0.4 inch. The same powder grains must not be poured twice to determine MPD since the level obtained would be affected by grains broken in previous pourings. For any weight of charge fired, PPD is determined from the PPD line and the charge is poured to that level.

4.3 Ballistic measurements. - Muzzle velocity and maximum breech pressure measurements are required for each round in this test except for conditioning rounds. Appropriate adjustment of the powder level is required for presence of pressure gages. Such adjustments may be determined empirically.

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soluble in ether (see 4.4), lead content (see 4.4), nitrogen

4.4 Firing schedule. - The order of firing below is to be followed in two guns. When the powder being tested is a cordite N powder, the cordite master powder must be used. When the powder being tested is either of the pyro (flashing or flashless) or the M6 (flashing or flashless) type then the M6 master powder is to be used. If at any time in the firing program a pressure of 20 tons per square inch (copper) is exceeded, firing should cease. Approximately 100 pounds of the test powder and 70 pounds of the master powder will ordinarily suffice to complete the firing schedule outlined below in two guns. 3"/50 caliber guns in which fewer than 1500 rounds have been fired are recommended for ballistic testing. Unnecessary over-service firing is to be avoided in guns used for ballistic testing.

Condition	Rd. No.	Powder	Charge	Remarks
1	1	Master	Service	Conditioning Round.
2	2,3,4	"	"	Determine mean velocity and pressure. Assigned master powder pressure minus this mean pressure is to be algebraically added to test powder pressures.
3	5	Test	3.0 lb.	Conditioning Round.
4	6,7,8	"	3.0 lb.	Determine mean velocity and pressure. Assume 490 f/s increase in velocity per pound of powder and compute charge C_o required to obtain mean velocity obtained in condition 2.
5	9	"	$\frac{3.0+C_o}{2}$	Probing Round. Check for excessive pressure.
6	10,11,12 13,14	"	C_1	Estimate service charge C_1 on basis of C_o and results of condition 5. (Round 10 may be considered a second probing round if charge C_1 appears to be grossly in error. In this case charge C_1 is to be re-estimated, say C_1', and five rounds would be fired with charge C_1'.) Compute mean velocity and maximum pressure. If mean condition 6 pressure corrected for pressure of the day does not exceed 18.0 tsi, fire:
7	15	"	C_2	Charge C_2 is calculated from firing in conditions 4, 5 and 6 to yield 100 feet per second greater velocity than condition 2.

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Condition	Rd. No.	Powder	Charge	Remarks
8	16	Master	Service	Conditioning Round.
9	17,18,19	"	"	Compute mean master powder velocity and pressure from results of conditions 2 and 9.

Repeat the entire program in the second gun. If more than one lot of powder is to be tested on one day in the same gun, Condition 9 will serve as Condition 2 for the second powder lot.

4.5 Application of Specifications. - The maximum and minimum acceptable service charges are listed in the table of 3.1. On the basis of the results of firing in Conditions 4, 5, 6 and 7, the charge is determined separately for each gun (by fitting a straight line to the velocity vs charge weight results) which is most likely to yield the mean velocity of Conditions 2 and 9. Service charge is taken as the average of the individual gun results. Exception: When testing any pyro powder or M-6 flashless powder, the M-6 flashing master powder is used and the type difference between test and master powder may affect the charge determination. The final service charge, W, is computed by:

$$W = \frac{\Delta V_o W_o - 1/2\,\Delta V_n (W_n + W_o)}{\Delta V_o - \Delta V_n}$$

Where: ΔV_o = 2700 f/s minus master powder velocity in the more worn gun

W_o = test powder service charge determined in the more worn gun

ΔV_n = 2700 f/s minus master powder velocity in the newer gun

W_n = test powder service charge determined in the newer gun.

A propellant powder for which the service charge derived in this way falls outside the limits cited fails the ballistic test. The maximum breech pressure expected for the service charge of the test powder (linearity of the pressure vs charge relationship is not assumed) corrected for the pressure of the day obtained from firing under Conditions 2 and 9 in the two guns is required not to exceed the values listed in the table of 3.1. The mean deviations (average absolute difference between individual results and the mean) are computed for the maximum pressures and muzzle velocities of the five rounds of Condition 6 averaged for the two firings. These values must also not exceed the values listed in the table of 3.1. If the results of the test fall outside the specifications in any one of the four respects listed, the test powder is regarded as failing ballistic test.

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4.6 Retest. - A propellant failing the ballistic test may be retested in the same guns with respect to the failing attribute. Failure of the powder to pass the retest is considered sufficient grounds for rejection of the lot.

5. PREPARATION FOR DELIVERY

(Not applicable in this specification.)

6. NOTES

6.1 Uses of powder types. - Pyro, M6 and Cordite N are all satisfactory propellants for the "slow-fire", hand-rammed, 3"/50 gun. These propellants may also be used in the "rapid-fire", power-rammed, 3"/50 gun, but Cordite N is preferred. When rounds are specified as "flashless", 1.0 ± .3% of potassium sulfate should be added to the Cordite N and 2.0 ± .3% to the pyro and M-6 powder. In "non-flashless" rounds the potassium sulfate should be omitted.

6.2 Safety precautions. - Since ballistic propellant testing involves firing an unproved component, reasonable safety precautions, such as providing proper shelter for personnel whenever test powder rounds are fired, should be observed.

Note. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplie the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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